Krish Kochar

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Education

University of Waterloo Sep 2022 - Apr 2027

Bachelor of Computer Science (AI Specialization) ***GPA: 4.0***

Selected Coursework: Algorithms, Operating Systems, Distributed Systems, Machine Learning, Computer Security

Experience

Incoming Software Engineering Intern @ Atlassian

Super.com May 2025 - Aug 2025

Software Engineering Intern

- Engineered **Go** backend services for booking, payment, & refund workflows, achieving **<50ms p95** service latency
- Designed an async webhook-based refund dispute pipeline with idempotency guarantees, ensuring correct semantics under retries, duplicate events, & partial failures, reducing resolution turnaround time by **30%**
- Integrated a third-party fraud service into the payments API with payments teams, reducing fraudulent bookings by **25%**
- Drove production triage, resolving **5+** critical incidents & introducing debugging workflows that reduced MTTR by **50%**

OTTO Motors by Rockwell Automation Jan 2025 – Apr 2025

Software Developer Intern

- Developed a multi-fleet interoperability system in **C++**, **Python**, & **ROS2**, improving coordination across **500+** robots
- Redesigned **networking architecture** by migrating from CycloneDDS to Zenoh, reducing communication latency by **25%** & increasing throughput by **30%**, validated with benchmarks & a production-ready **PoC** for fleet deployment
- Iteratively tuned **QoS** settings to reduce packet loss in stress tests, boosting E2E communication across distributed nodes
- Replaced polling with an **event-driven ROS2** topic for real-time robot tracking, reducing CPU usage by **20%**

Ford Motor Company May 2024 - Aug 2024

Platform Software Developer Intern

- Deployed a real-time Vision Compute Service in **C++** with **OpenVX** & **OpenCV**, accelerating sensor data processing & optimizing multi-threaded workloads for high-performance execution on embedded hardware
- Integrated **Gcov** into **Jenkins** CI pipelines, enabling **80%** functional test coverage visibility & unblocking QA workflows
- Built **Python** scripts for release validation and code-size tracking, surfacing inefficiencies that cut firmware size by **20%**
- Debugged & resolved **critical memory defects** with Valgrind, eliminating leaks & stabilizing production firmware

Publicis Sapient May 2023 - Aug 2023

Software Engineering Intern

- Implemented **Go** APIs for an employee management portal with **300+** DAU, supporting CRUD & auth workflows
- Optimized data-fetching paths by introducing **GraphQL** aggregation, reducing API response times by **30%**

Research

University of Waterloo & Vector Institute Sep 2024 - Dec 2024

Undergraduate Research Assistant, Prof. Xi He

- Generated a synthetic dataset with **<10%** divergence from real data, using a ClavaDDPM **diffusion** pipeline in **Python**
- Benchmarked data privacy via black-box **Membership Inference Attacks** to assess the risks of sharing synthetic data

Projects

Aura - Code Review Buddy | *Python, FastAPI, Cerebras, ChromaDB, Ollama*

- Built a GitHub browser extension that generates AI code summaries in **<1s**, helping review PRs faster with better context

RaIInet | *C++, Xlib, UML*

- Designed a **C++** board game with the **MVC** architecture, using **OOP** and **SOLID** principles for modularity & scalability

Technical Skills

Languages: C, C++, Go, Rust, Python, SQL, TypeScript, JavaScript, Bash

Systems & Tools: Docker, Kubernetes, Helm, Jenkins, Git, Redis, Datadog, Google Test

Frameworks & Libraries: ROS2, OpenCV, FastAPI, React, Node.js, Pandas, NumPy, scikit-learn, TensorFlow, PyTorch